THE FIVE STANDING PROPERTIES HOLD FOR THE COMPARISON OF THE LOCALLY COUNTABLY GENERATED SITE WITH ITS ZARISKI COUNTERPART

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the weak Serre verification through the four conditions, the four cited properties, and the explicit construction refining a qc covering by a surjective proper map.

Editorial note. The label lemma-compare-qc-zar names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-compare-qc-zar –start 7985 –end 8046 –local-id 3556 returned RESULT=OK (L3556 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement refers to the five labelled items of the standing comparison situation; these are kept as external pointers with their canonical Stacks tags.

Editorial note. The statement and proof are copied from the pinned chapter file.

Allowed standard prerequisites (editorial).

- Characterization of weak Serre subcategories. Homology, the lemma listing conditions (1)-(4) characterizing a weak Serre subcategory of an abelian category. Supplies the four conditions that are checked to show that the full subcategory of sheaves of the form pi_X^{-1}F is a weak Serre subcategory.
- The pullback of the comparison is exact and fully faithful. Same chapter, the lemma collecting the true statements about the Zariski comparison, in particular the exactness and full faithfulness of pi_X^{-1} and the exactness of pi_{X,*} on short exact sequences of sheaves in the image. Supplies conditions (1)-(3) of the weak Serre criterion and the pushforward exactness used to check the fourth condition.
- Basic properties of the locally countably generated site and properness of base changes. Same chapter, the lemma recording the basic properties of the site, together with the topological characterization of proper maps and the base change of separated maps. Supplies property (item-base-change-P), because base change of a proper map is proper.
- The sheaf property for pullbacks along the projection and the compatibility of the class with proper maps. Same chapter, the lemma describing the pullback along the projection of the site, and part (7)(b) of the lemma

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collecting true statements about the Zariski comparison. Supply property (item-A-sheaf) and property (item-A-and-P) respectively.
- A surjective proper map is a qc covering. Same chapter, the lemma that a surjective proper morphism of topological spaces is a qc covering. Supplies the last step of the construction refining a qc covering by a proper map.

Source reference. Original lemma 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Prerequisite pointer invoked at proof line(s) 8006; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ7: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6862–6862. Prerequisite pointer invoked at proof line(s) 7998; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ6: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6857–6857. Prerequisite pointer invoked at proof line(s) 7997; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ4: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6849–6849. Prerequisite pointer invoked at proof line(s) 7995; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6868–6868. Prerequisite pointer invoked at proof line(s) 7999; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6853–6853. Prerequisite pointer invoked at proof line(s) 7996; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7474–7474. Prerequisite pointer invoked at proof line(s) 8017; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DCU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7770–7770. Prerequisite pointer invoked at proof line(s) 8008; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09X3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7675–7675. Prerequisite pointer invoked at proof line(s) 8027; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09X5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7585–7585. Prerequisite pointer invoked at proof line(s) 8043; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6841–6841. Prerequisite pointer invoked at proof line(s) 8000; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CY4: sources/stacks-chow/tags.txt, source lines 16229–16229. Prerequisite pointer invoked at proof line(s) 8020; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08YD: sources/stacks-chow/tags.txt, source lines 11336–11336. Prerequisite pointer invoked at proof line(s) 8040; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 005R: sources/stacks-chow/tags.txt, source lines 219–219. Prerequisite pointer invoked at proof line(s) 8019; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider the comparison morphism $\epsilon : LC_{qc} \to LC_{Zar}$. Let $\mathcal{P}$ denote the class of proper maps of topological spaces. For X in LC_{Zar} denote $\mathcal{A}'_X \subset Ab(LC_{Zar}/X)$ the full subcategory consisting of sheaves of the form $\pi_X^{-1}\mathcal{F}$ with $\mathcal{F}$ in $Ab(X)$. Then (0EZ4), (0EZ5), (0EZ6), (0EZ7), and (0EZ8) of Situation 0EZ3 hold.*

Proof. We first show that $\mathcal{A}'_X \subset Ab(LC_{Zar}/X)$ is a weak Serre subcategory by checking conditions (1), (2), (3), and (4) of Homology, Lemma 0754. Parts (1), (2), (3) are immediate as π_X^{-1} is exact and fully faithful by Lemma 0DCU part (3). If $0 \to \pi_X^{-1}\mathcal{F} \to \mathcal{G} \to \pi_X^{-1}\mathcal{F}' \to 0$ is a short exact sequence in $Ab(LC_{Zar}/X)$ then $0 \to \mathcal{F} \to \pi_{X,*}\mathcal{G} \to \mathcal{F}' \to 0$ is exact by Lemma 0DCU part (2). Hence $\mathcal{G} = \pi_X^{-1}\pi_{X,*}\mathcal{G}$ is in $\mathcal{A}'_X$ which checks the final condition.

Property (0EZ4) holds by Lemma 09WZ and the fact that the base change of a proper map is a proper map (see Topology, Theorem 005R and Lemma 0CY4).

Property (0EZ5) follows from the commutative diagram (5) in Lemma 0DCU.

Property (0EZ6) is Lemma 09X3.

Property (0EZ7) is Lemma 0DCU part (7)(b).

Proof of (0EZ8). Suppose given a qc covering $\{U_i \to U\}$. For $u \in U$ pick $i_1, \ldots, i_m \in I$ and quasi-compact subsets $E_j \subset U_{i_j}$ such that $\bigcup f_{i_j}(E_j)$ is a neighbourhood of u. Observe that $Y = \coprod_{j=1,\ldots,m} E_j \to U$ is proper as a continuous map between Hausdorff quasi-compact spaces (Topology, Lemma 08YD). Choose an open neighbourhood $u \in V$ contained in $\bigcup f_{i_j}(E_j)$. Then $Y \times_U V \to V$ is a surjective proper morphism and hence a *qc* covering by Lemma 09X5. Since we can do this for every $u \in U$ we see that (0EZ8) holds. □